Song Recommendation

Data Mining Final Project Presentation
Group 11

Apitsada (Pearl) Ruknapapong, Daeun Ji, Kavya Bhat,
Chi Nguyen, Shubham Kumar

Agenda

- Project Objective
- Data Overview
- Methodology
- Findings
- Conclusions

"The best songs are the ones that come to you at the right moment, when you need them the most."

– Unknown

Project Objective

1. **Personalized Song Recommendation System**:

   Recommend songs that align with **individual preferences** based on both **audio features** and **text-based responses (comment sentiments)** from listeners.

2. **Multi-label, Multi-class Classification of Songs**:

   Predict multiple **sentiment labels (derived from text-based responses)** for each song based on its **audio features** and **lyrical content**.

Data Overview

Primary Dataset:

- **Spotify Song Attributes**: This includes metadata for songs such as artist name, song name, album, danceability, energy, loudness, Speechiness, Acousticness, Instrumentalness, Liveness, Valence, Tempo etc., along with lyrics.

Secondary Dataset:

- **YouTube Comments**: For each song, YouTube comments were scraped (via YouTube API). The comments were categorized into different 'categories' by **using LLMs and prompt-engineering**. Categories such as:
  - **Humor and Memes**
  - **Appreciation and Praise**
  - **Words of Empathy (Deeply emotional)**
  - **Personal Stories and Experiences**

Data Columns:

- **Spotify Data**: Track ID, artist name, song name, album, duration, energy, danceability, and more.
- **YouTube Comments**: Comments were categorized into emotional response categories and song topics like "Love and Relationships", "Celebration and Fun", "Spiritual and Religious" etc.

Data Overview

Column	Description
Spotify ID	Track ID on Spotify
Artist	Artist name
Spotify URL	Track URL on Spotify
Track	Song name
Album	Album name
Album_type	Album Type (e.g., album, single)
Duration_ms	Duration of song in milliseconds
Song Attributes	Score for song attributes (Danceability, Energy, Key, Loudness, Speechiness, Acousticness, Instrumentalness, Liveness, Valence, Tempo)

Data Overview

Column	Description
YouTube URL	YouTube URL for the song
YouTube Comment Categories	Number of times that listeners' emotional responses appear in the category (Humour & Memes, Appreciation & Praise, Words of Encouragement, Words of Empathy, Personal Stories & Experiences, Nostalgia & Memories) Meaningful/reliable classifications occurred for: Humour & Memes, Appreciation & Praise, Words of Empathy (Deeply Emotional), Personal Stories & Experiences
Title	Youtube video title
YouTube Video ID	Video ID on YouTube
Lyrics	Song lyrics
Tokens	Word tokens from the lyrics
Dominant Topic Labels	Topic from lyrics (Love & Relationships, Self-Reflection and Personal Struggles, Social and Political Themes, Celebration and Fun, Philosophical and Existential, Storytelling and Narrative, Escape and Fantasy, Spiritual and Religious, Cultural and Lifestyle, Fun and Humor)

Lyric topics using LLMs and Prompt-engineering:

- Love & Relationships
- Self-Reflection and Personal Struggles
- Social and Political Themes
- Celebration and Fun
- Philosophical and Existential
- Storytelling and Narrative
- Escape and Fantasy
- Spiritual and Religious
- Cultural and Lifestyle
- Fun and Humor

Feature meanings:

1. **Speechiness**: Measures the presence of spoken words in a track.
2. **Energy**: Measures the intensity and activity of the track.
3. **Danceability**: Describes how suitable a track is for dancing.
4. **Valence**: Measures the musical positiveness conveyed by the track.
5. **Loudness**: The overall loudness of a track in decibels (dB).
6. **Acousticness**: Measures the degree to which a track is acoustic.
7. **Duration_ms**: The duration of the track in milliseconds.
8. **Tempo**: The tempo of the track in beats per minute (BPM).
9. **Liveness**: Describes the presence of a live audience in the recording.
10. **Instrumentalness**: Measures the likelihood that the track is instrumental
11. **Key**: The musical key of a track (e.g., C Major, A Minor) represents the tonal center and the scale upon which the song is based. It defines which notes and chords are most prominent throughout the track..

Data Overview

YouTube Comment Categories

Lyrics Topics

Raw Data

- Song attribute & lyrics: 955,320 rows x17 cols
- Spotify & YouTube URL: 18,862 x 28 cols

Match by Spotify Track ID

Cleaned Data

- Filtered English songs
- 3,234 rows x 38 cols

Methodology

1. **Data Preprocessing**:
    - Filtering English songs and handling missing values.
    - Merging datasets from Spotify and YouTube based on song ID.
2. **Clustering**:
    - Used **Agglomerative Clustering** on both **audio features** and **comment categories** to form meaningful clusters based on similarity.
    - This clustering helps identify groupings of songs with similar emotional tones or audio characteristics.
3. **Recommendation System**:
    - Songs are recommended based on their **Euclidean distance** to a user-selected song.
    - The system compares audio features and emotional responses, returning the top 5 similar songs based on the calculated similarity.
4. **Multi-label, Multi-class Classification**:
    - We also experimented with multi-label multi-class classification to predict audience responses using the final dataset.
5. **Evaluation**:
    - The model performance was evaluated using **Subset Accuracy** and **Micro F1-Score** across different labels

Clustering approaches tried and findings for comments and spotify metadata.

Following clustering algorithms were tried:-

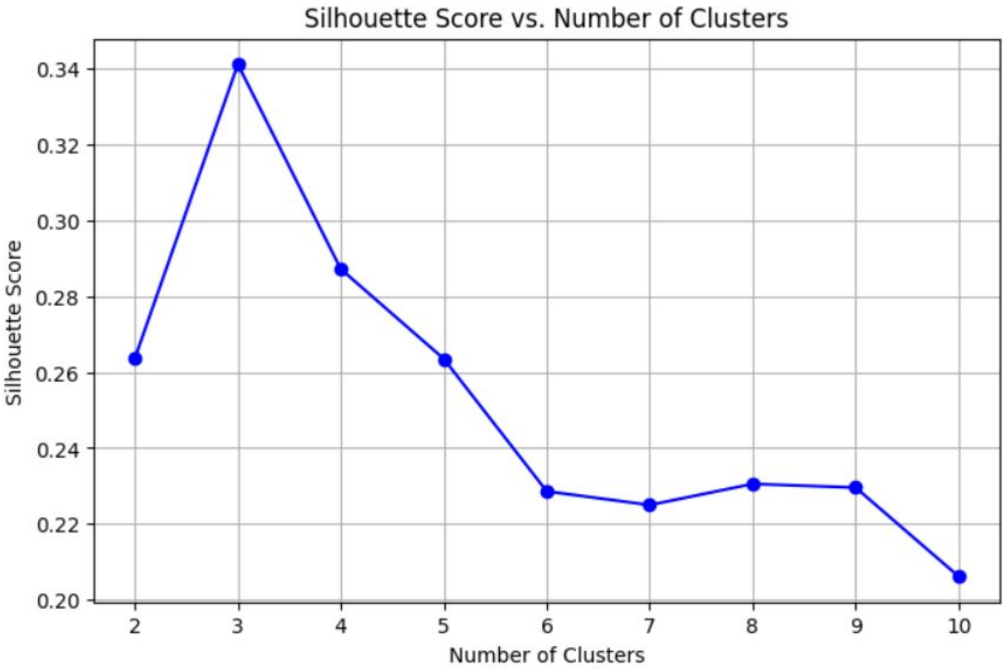
1. DB-Scan:
   a. This resulted in various small clusters indicating high variability in the data.

2. K-Means:
   a. The scree plot (wcss vs total-clusters) never completely plateaued which further affirmed the high complexity/variability present in the data.

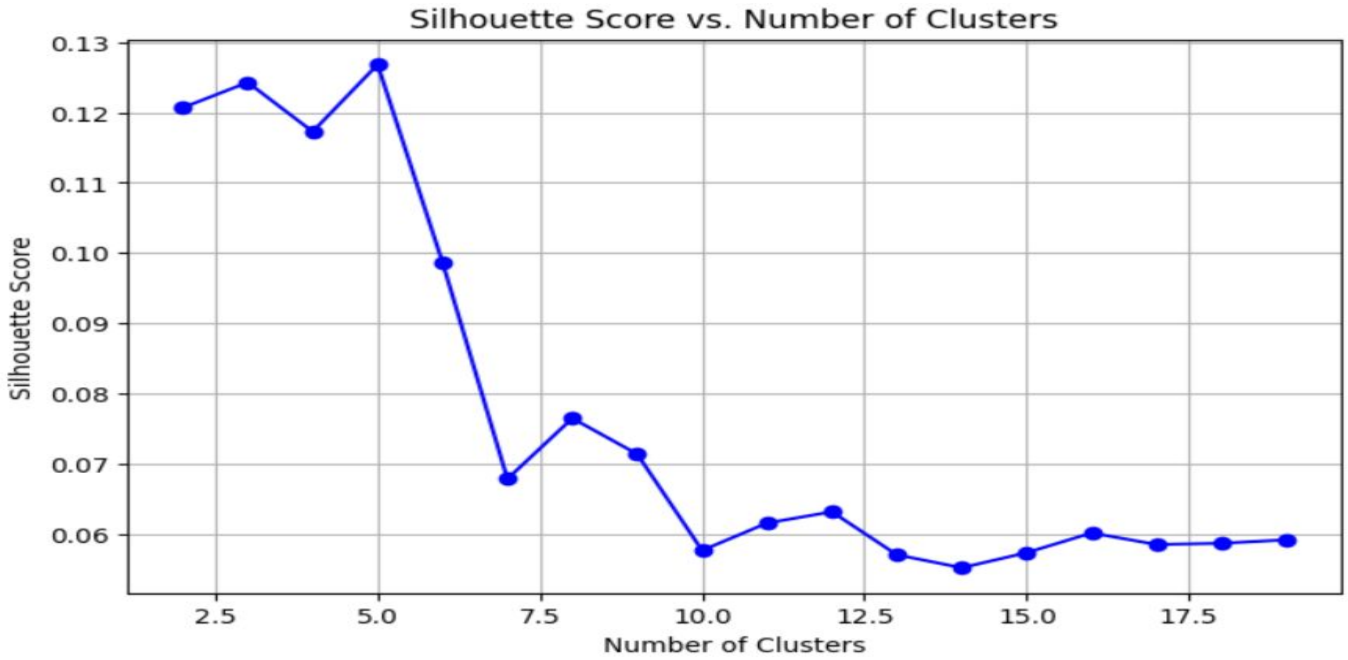
3. Agglomerative clustering:
   a. Ideally suited to hierarchical data, this can be used for data with no clear hierarchies as well.
   b. This provided meaningful insights into the clusterings that the data can form using ‘ward’ linkage and ‘euclidean’ distance metric.

Note: Lack of useful clusters, or detection of various clusters, using DB-Scan and K-Means could also be attributed to the specific sample picked for the analysis, and these approaches might have performed better for a different sample.

Visualisations (Agglomerative on comments) - 1



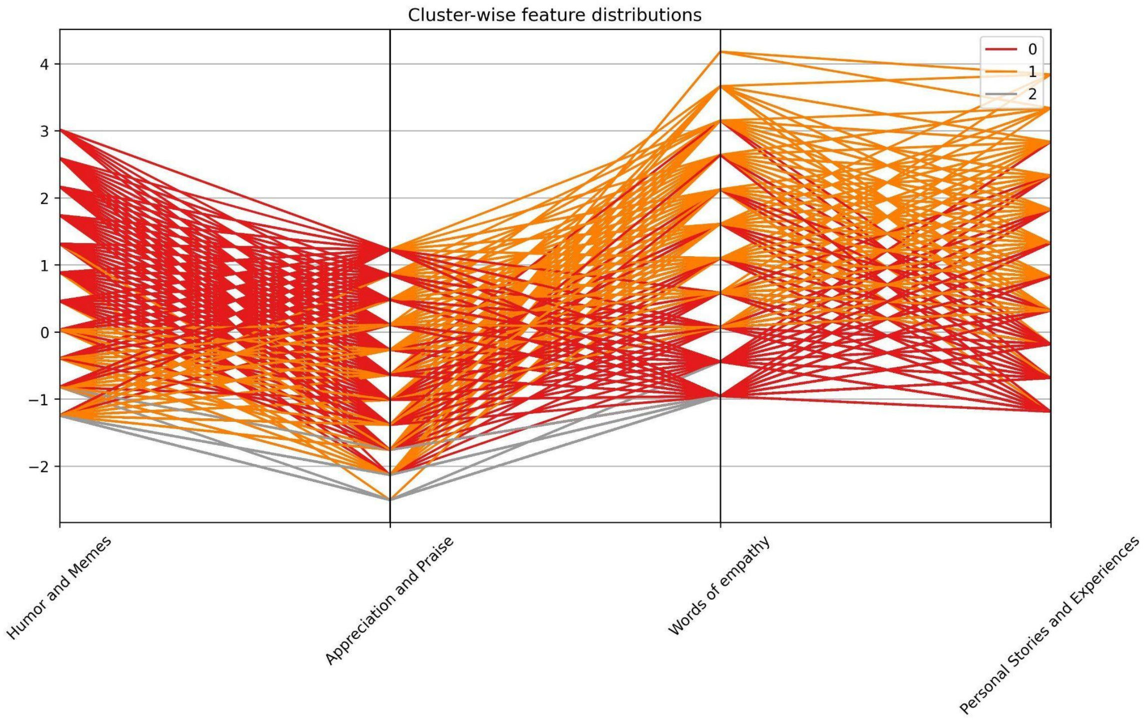
Visualisations (Agglomerative on spotify) - 2



Findings - Clustering

Parallel-Pair plot:-

- Cluster 0 (red) represents higher concentration of **Humor and Memes** related comments
- Cluster 1 (orange) represents higher concentration of **Words of empathy** (deeply emotional) related comments as well **Personal Stories and Experiences**
- **Appreciation and Praise** has high concentration in both red and orange clusters.
- Cluster 2 represents all records which have relatively lower concentrations of all types of comments



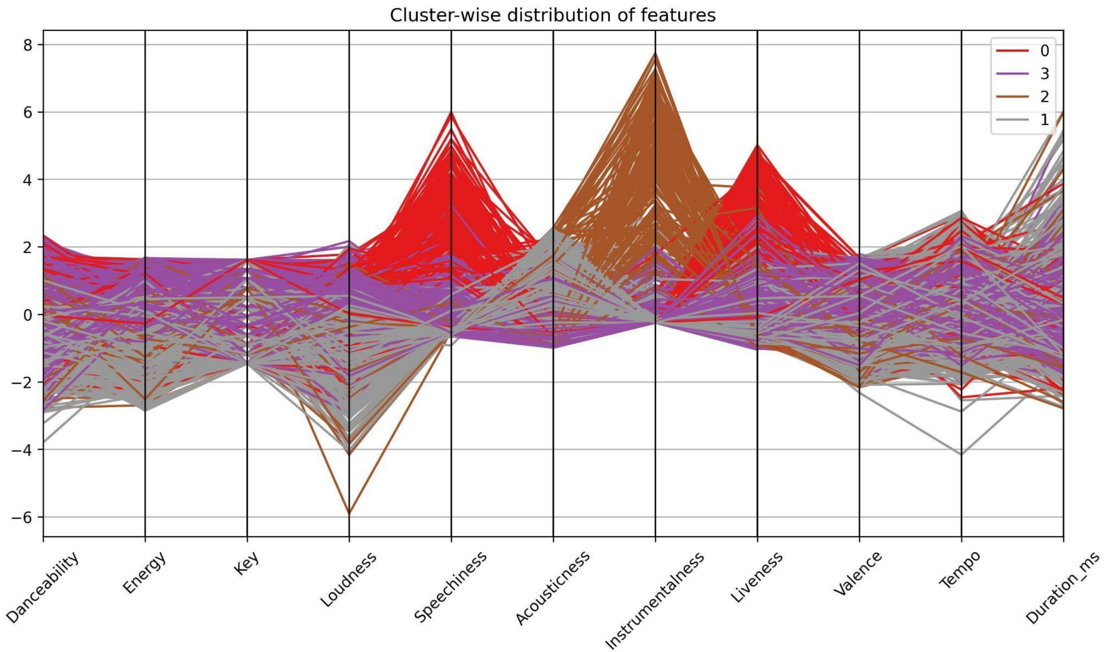
Findings - Clustering

- Cluster 0 (red) represents high concentrations of songs with **Speechiness** and **Liveness.** This could imply live performances.

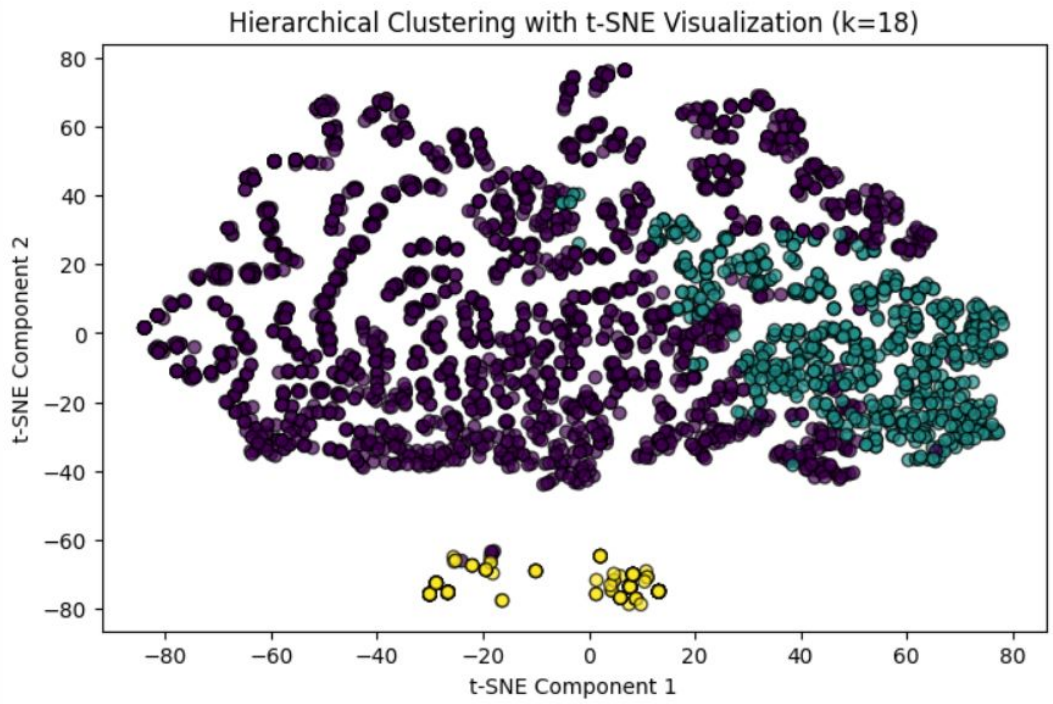
- Cluster 1 (gray) represents highest concentration of **Acousticness,** and simultaneously has songs with relatively low energy and loudness.

- Cluster 2 (brown) represents high concentration of **Instrumentalness**

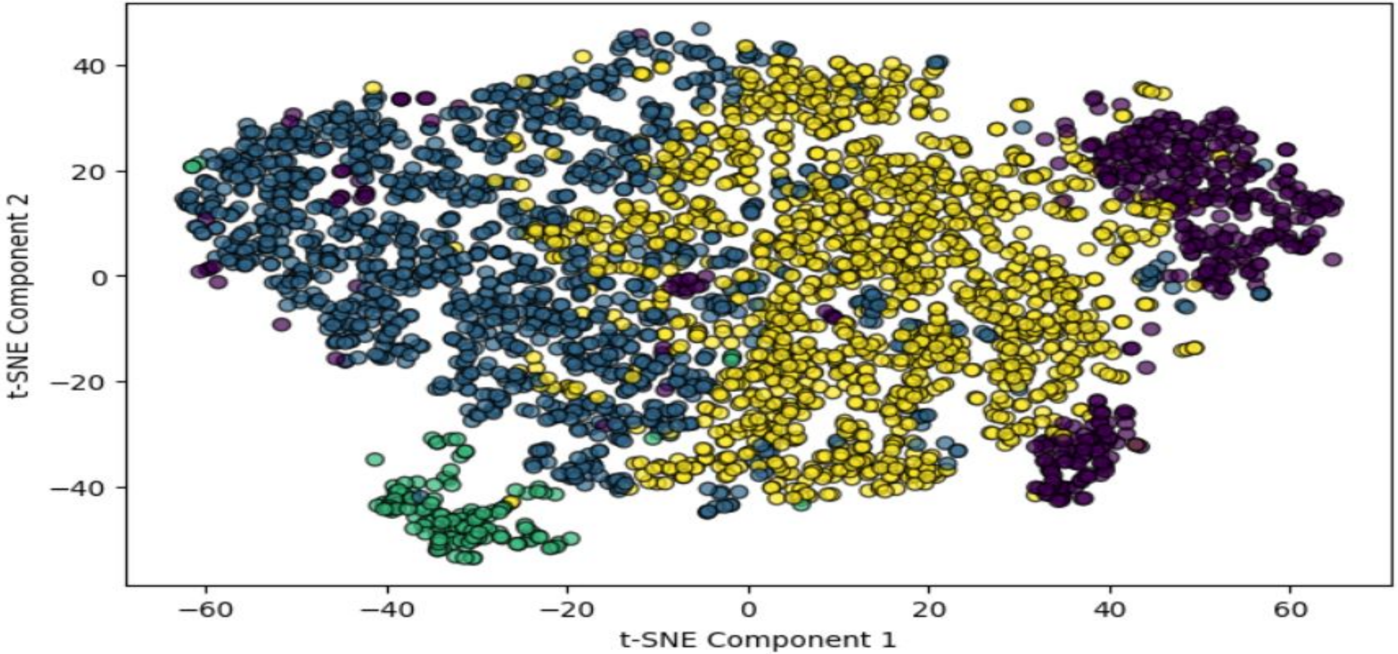
- Cluster 3 (purple) represents **balanced distribution** across all clusters.



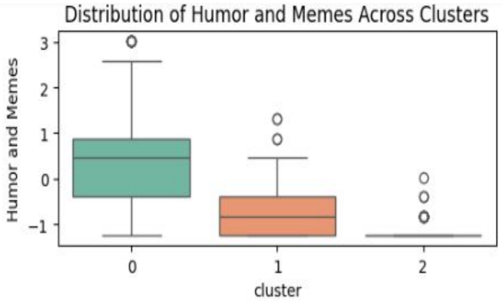
Visualisations (Agglomerative on comments) - 2

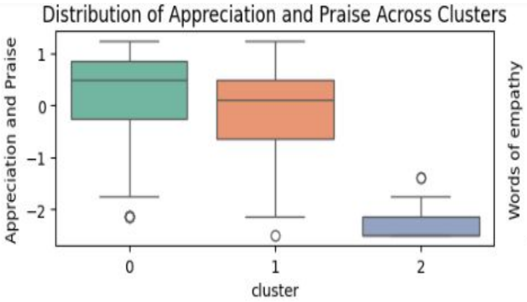


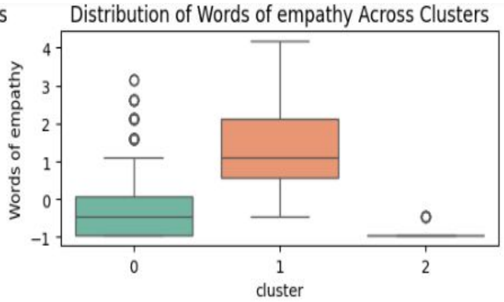
Visualisations (Agglomerative on spotify) - 2

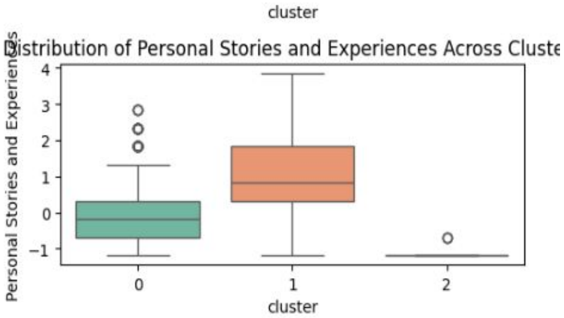


Cluster-wise box plots for different attributes:

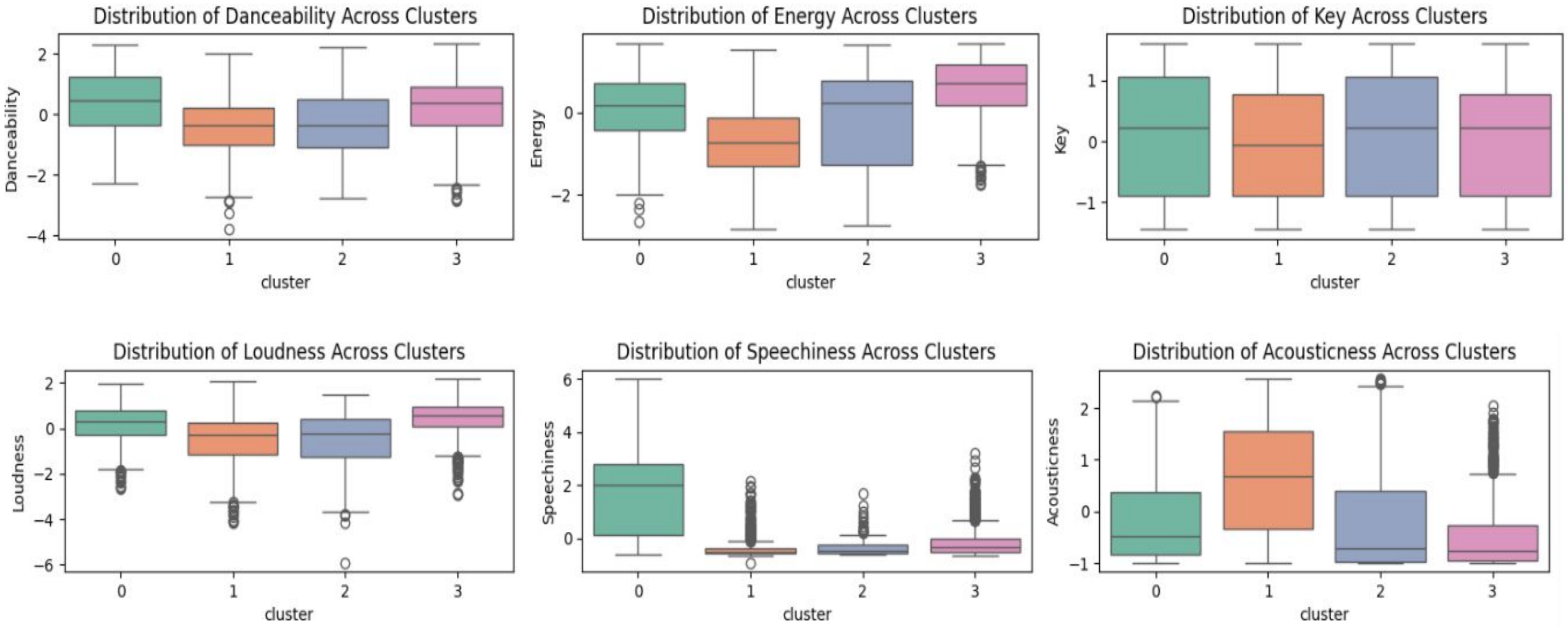




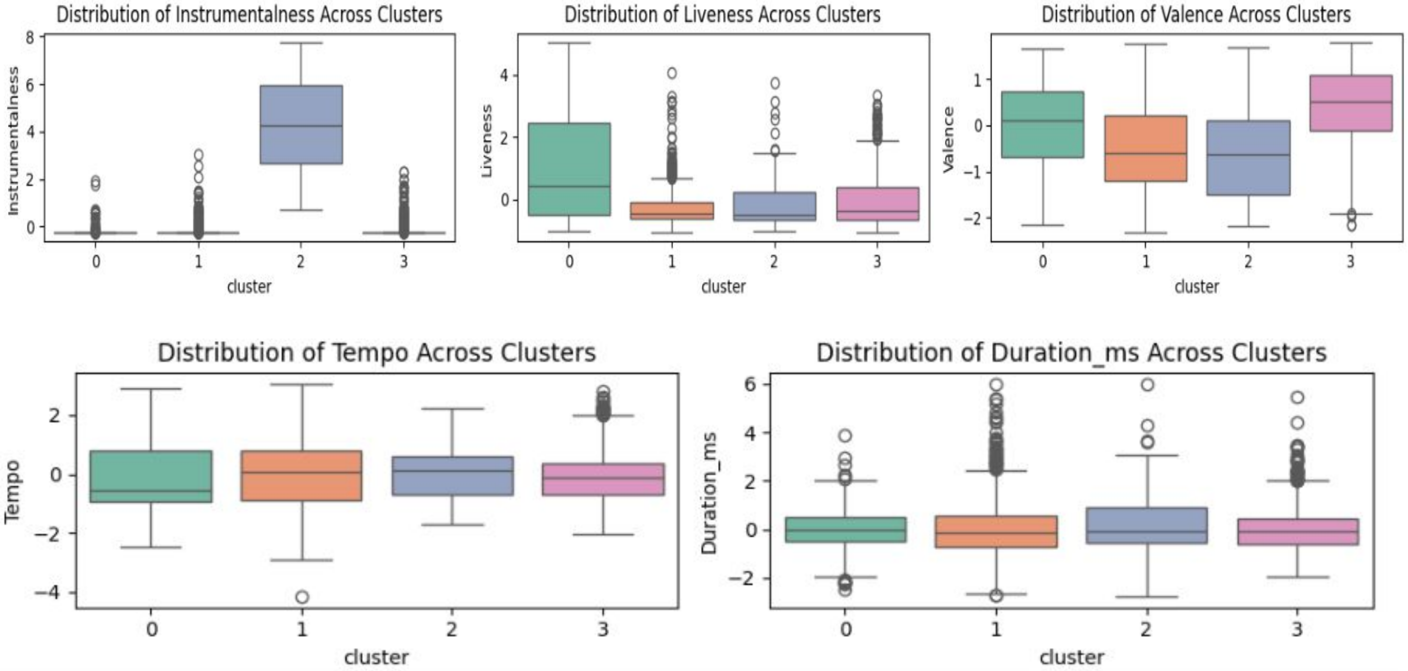




Cluster-wise box plots for different attributes - 1:



Cluster-wise box plots for different attributes - 2:



Recommendations based on comments

```
merged_df['order'] = pd.Categorical(merged_df['spotify_id'], categories=spotify_ids_found_based_on_comments, ordered=True)
merged_df = merged_df.sort_values('order')
merged_df[ ['Artist', 'Track', 'Humor and Memes', 'Appreciation and Praise', 'Words of empathy', 'Personal Stories and Experiences', 'spotify_id', 'clust
```

[44]:

	Artist	Track	Humor and Memes	Appreciation and Praise	Words of empathy	Personal Stories and Experiences	spotify_id	cluster
7	James Taylor	Carolina in My Mind	1	8	7	9	2T5Ch09nefwckOu5NQvjlk	1
2	Korn	Falling Away from Me	4	6	7	7	2F6FfZ4w8z3eJpSxPotVO5	1
4	Franco De Vita	Tú de Qué Vas	0	8	8	7	66iygyOSvvoQQsKJ1vEXfT	1
1	Juanes	Para Tu Amor	2	7	6	8	2quTytCTP9kKLvpveyO1mt	1
6	Rascal Flatts	I Won't Let Go	1	7	9	6	1T1C9VVQ1lb60WxWZx7KoG	1
5	Leo Dan	Qué Tiene la Niña	1	9	6	6	68tjkySJsysjABJ5HATZLR	1
9	Godsmack	Under Your Scars	1	9	6	6	60SjEdocmsrAk4hQUu7qZi	1
0	Fleetwood Mac	Gypsy	1	8	5	8	19Ym5Sg0YyOCa6ao21bdoG	1
3	Van Morrison	Brown Eyed Girl	0	6	7	9	3yrSvpt2l1xhsV9Em88PuI	1
8	Jarabe De Palo	Eso Que Tú Me Das	0	10	7	5	40tFJtuES1da2fg0OurUdI	1

[31]:

```
df_final_aggregated[df_final_aggregated['Track'] == 'Chasing Cars' ][ ['Artist', 'Track', 'Humor and Memes', 'Appreciation and Praise', 'Words of empathy
```

[31]:

	Artist	Track	Humor and Memes	Appreciation and Praise	Words of empathy	Personal Stories and Experiences
2531	Snow Patrol	Chasing Cars	2	4	5	5

Comments corresponding to one of the recommended songs - Carolina in my Mind

Refer to spotify_comments_recent_and_labels.csv and filter using "Track" column:

comment
This song just destroys me. I'm a native North Carolinian and graduate of North Carolina. Jame's Taylor's father was a professor and Dean at Carolina and he grew up there. For me this song will always be associated with those first few days back on c
This song brings tears to my eyes. So beautiful. People used to make fun of me for loving James Taylor. He's such a great man.
Song reminds me of my Dad.. Wish I could turn back time and spend one more day with you Dad..Rest in eternal peace my old friend.. I love you always x
Beautiful and calming.   Greetings from the beautiful blue ridge mountains of Asheville in western North Carolina!!!!!!!!
Sitting on my patio with a glass of wine...sundown...during the Coronavirus situation...but I'm still thankful.
The thing that appeals to me about him is that he sings it straight from his heart, with that beautiful glint in his eyes , on how much he misses his home and his yearning to go there despite living his dream with the 'holy host' (the beatles) ... Anyone who
This was my grandmother's favorite song. Her favorite place to be was up in the mountains in North Carolina. She died a few months ago, and we had her funeral up in the mountains today, on what would have been her 79th birthday. As she requested,
This song took me through many years of international living. And now Iâ€™M BACK. You can take the girl out of Carolina, but you can NOT take the Carolina out of the girl. Virtual high five to all of you ðŸ'ŠðŸ¼
Saw him in Paris last spring ; At intermission instead of going backstage he sat down on the stage and spoke for a 1/4 hour with fans...In French !!
Played this song on my Dads memorial video along with Fire and rain hard to listen to but makes me happy when I hear it. Love you Dad

Insights from the recommended songs:

It can be clearly seen that comments for the recommended songs have higher concentrations of ‘words of empathy’ and ‘personal stories’ comments which fit the criteria for cluster 1 that the song ‘Snow Patrol’ belongs to.

The comments can indeed be considered **deeply emotional (words of empathy) as well as accounts of personal experiences**.

One of the comments (8th comment) has an **element of humor** as well.:

“You can take the girl out of Carolina. But you can NOT take Carolina out of the girl”

Comparing that to the comments under reference song - Chasing Cars

comment
Who's here for no other reason at all apart from they wanted to hear a great song ?
â€œItâ€™s a beautiful day to save livesâ€
When my wife was dying of cancer, we used to lay on our bed listening to this and other songs which we had memories attached to. Few weeks later she went into a hospice and 27 days later she died. Now I lay here thinking and thinking and hoping tha
When I had breast cancer I decided if I didnâ€™t make it I wanted this song played at my funeral. Instead 13 years later it was the first dance at my wedding. Will always be one of my favourites. Thank you Mr Lightbody!
A message to the future generations. Don't let this song die.
My grandpa was a car sales man, he wanted this song in his funeral. He died this morning at 4:44 AM. He helped me through every part of my life, financially and emotionally, I told him every secret Iâ€™ve ever had, he never told anyone. Rest In Peace G
I held my husband in my arms while he took his last breaths. For 21 years we fell asleep holding hands. He did not die alone in his own arms. Five years ago and the moment is in a special safe place in my heart. His last words were I don't know how heav
â€œHi everyone, itâ€™s a beautiful day to save livesâ€ - Derek shepherd
Anyone still here with me crying and listening to this song now? This song hits different whenever I listen to it.
My dad, the most stable person in my life and my best friend, died yesterday in corona. He was by himself in the hospital. I could not say goodbye to him. Listening to this song and crying, how i can live without him.

Findings - Recommender

Recommendation Basis:

- Uses **audio features** (e.g., danceability, tempo, energy) and **YouTube comments** (e.g., emotional response) for recommendations.

Clustering:

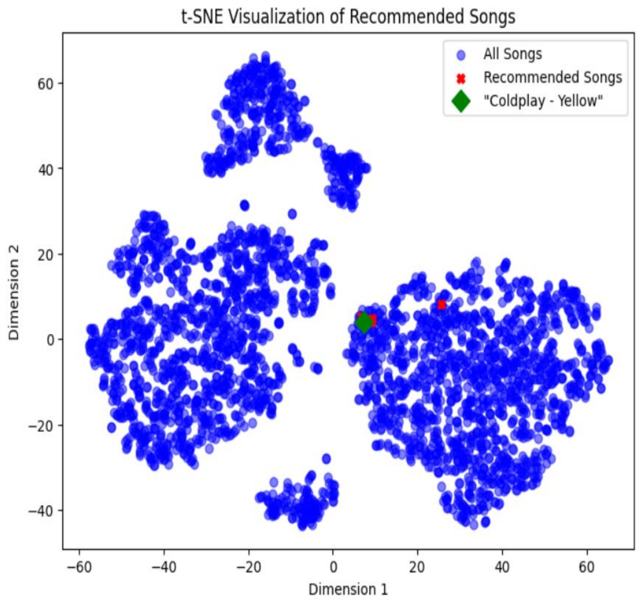
- Songs are clustered based on audio features and comments.
- 5 clusters are formed based on similarity in audio characteristics or emotional responses.

Similarity Calculation:

- **Euclidean Distance** and **Cosine Similarity** are used to calculate song similarity.
- Top 5 similar songs are recommended.

Findings - Recommender Example

Based on **"Coldplay - Yellow"**, songs with similar **audio features** or **emotional responses** are recommended.



Graph:

Blue dots: All songs.

Red Xs: Top 5 recommended songs based on similarity to "Coldplay - Yellow".

Green diamond: "Coldplay - Yellow".

Song Cluster: Cluster 3 (**balanced distribution** across all clusters.)

Top 5 Recommended Songs:

Sia - Unstoppable, Luis Miguel - La Incondicional, Bryan Adams - Heaven, Toby Keith - Made in America, Creed - My Own Prison

	Album	Title	spotify_id
200	This Is Acting (Deluxe Version)	Sia - Unstoppable (Official Video - Live from ...	1yvMUkIOTeUNtNWIWRgANS
275	Busca Una Mujer	Luis Miguel - "La Incondicional" (Video Oficial)	6F9yAYUaNbUhdlQyt5uZ3b
958	Reckless (30th Anniversary / Deluxe Edition)	Bryan Adams - Heaven	7Ewz6bJ97vUqk5HdkvguFQ
2736	Clancy's Tavern	Toby Keith - Made In America (Official Music V...	7Lmwj2fe8MpGXypOuLGO2C
2840	My Own Prison	Creed - My Own Prison	5vRPXm59z8ewWO6WiJHg3m

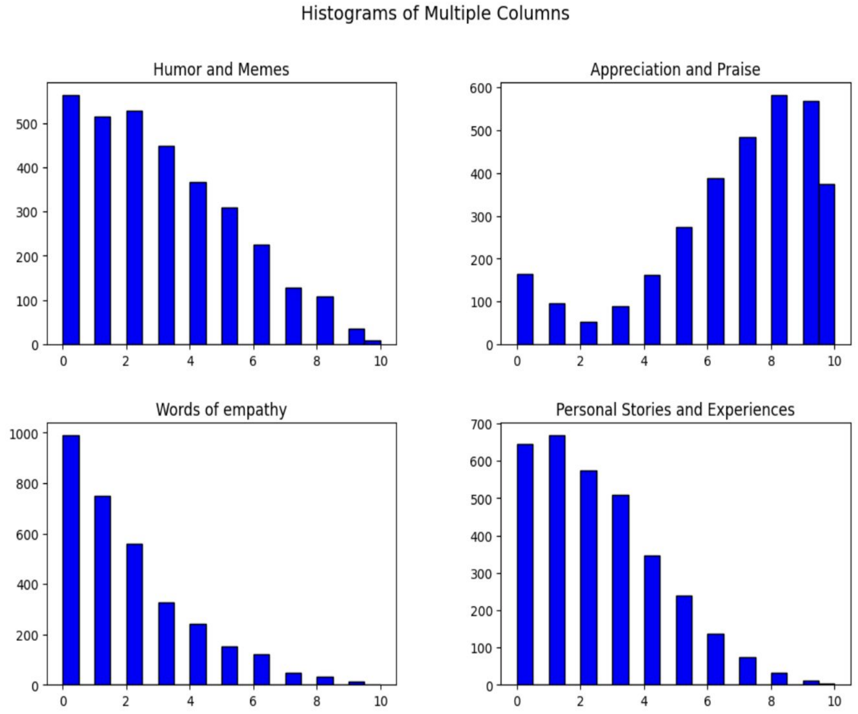
Findings - Multi-label/ Multi-class Classification

Multi-label Multi-class Classification:

- **Definition**: Predicting **multiple labels** for each sample. Each sample can have more than one label assigned simultaneously.
- **Example**: A song can have multiple labels, such as **"Humor"** (comment category) and **"Energy"** (Spotify attribute), assigned at the same time.

Multi-class Classification:

- **Definition**: Predicting **one label** for each sample, where each sample belongs to only one class or label.
- **Example**: A song can belong to only one main label, such as **"Humor"** (comment category) or **"Energy"** (Spotify attribute), but not both at the same time.



Preprocessing to create labels - methodology

Here are brief points about the methodology to convert comment-category counts to labels:

1. **Percentile Calculation**: Compute 25th, 50th, 75th, and 95th percentiles to understand value distributions.
2. **Thresholding for Label Creation**: Use median as a threshold to convert continuous values into binary labels.
3. **Binary Labeling Strategy**: Values above the median are considered positive instances (label=1).
4. **Multi-Label Aspect**: Allows multiple categories to be activated for a single sample.
5. **Potential Limitations**: May lose nuanced differences; threshold choice impacts performance.
6. **Data Interpretation**: Simplifies data but might lose original information.
7. **Modeling Implications**: Suitable for classification algorithms; performance depends on threshold alignment.

Models tried: 1. Random Forest

Model of Choice: Random Forest with MultiOutputClassifier

Why?

1. **Multi-Label Handling**: Effective for multi-label classification.
2. **Robustness**: Reduces overfitting risk.
3. **Feature Importance**: Provides insight into contributing features.

Interpretation of Results:

- **Subset Accuracy (0.1716)**: Indicates exact matches are rare. This is generally considered very challenging to achieve a good score for.
- **Micro F1-Score (0.5294)**: Shows moderate performance across labels in terms of precision and recall across all labels.

K-Fold Cross Validation and Hyperparameter Tuning

Best Parameters:

```
{
    'estimator__max_depth': 10,
    'estimator__max_features': 'sqrt',
    'estimator__min_samples_leaf': 5,
    'estimator__min_samples_split': 2,
    'estimator__n_estimators': 100
}
```

Best Score (on training set with CV): 0.5194652368502145

Subset Accuracy (on test set): 0.1901

Micro F1-Score (on test set): 0.5446

XGBoost

XGBoost Test Accuracy: 0.1932
XGBoost Micro F1-Score: 0.5379

Sample output: Original Data

Unnamed: 0	1334
spotify_id	35xilew5nalcetOeytaDFj
Love and Relationships	1
Self-Reflection and Personal Struggles	1
Social and Political Themes	0
Celebration and Fun	0
Philosophical and Existential	0
Storytelling and Narrative	1
Escape and Fantasy	0
Spiritual and Religious	0
Cultural and Lifestyle	0
Fun and Humor	0
Humor and Memes	0
Appreciation and Praise	1
Words of encouragement	5
Words of empathy	1
Personal Stories and Experiences	1
Nostalgia and Memories	8
Artist	Jay Chou
Url_spotify	https://open.spotify.com/artist/2elBjNSdBE2Y3f...
Track	還在流浪
Album	最偉大的作品
Album_type	album
Uri	spotify:track:35xilew5nalcetOeytaDFj
Danceability	0.13219
Energy	-0.431985
Key	-1.178061
Loudness	0.114041
Speechiness	-0.538549
Acousticness	-0.596542
Instrumentalness	-0.24107
Liveness	-0.488841
Valence	-1.845201
Tempo	1.558098
Duration_ms	0.424081
Url_youtube	https://www.youtube.com/watch?v=BSlMAJ7SkMA

Sample output: Predicted Data

```
Duration_ms                                       0.424081
Url_youtube            https://www.youtube.com/watch?v=BSlMAJ7SkMA
Title                  周杰倫 Jay Chou【還在流浪 Still Wandering】Official MV
id                                      35xilew5nalcetOeytaDFj
lyrics                 老書攤 和啤酒空罐 散落地上\n 報廢的回憶在路旁 有了銹斑\n 有某種無所謂 在這小鎮彌漫...
youtube_video_id                                   BSlMAJ7SkMA
video_id                                           BSlMAJ7SkMA
```

```
Prediction for the test point: [[0 0 1 1]]
Accuracy on test set: 0.1901081916537867
```

Feature importance

	Feature	Overall Importance
4	Speechiness	0.133070
1	Energy	0.096312
0	Danceability	0.087693
8	Valence	0.085752
3	Loudness	0.083356
5	Acousticness	0.081108
10	Duration_ms	0.081105
9	Tempo	0.075396
7	Liveness	0.070180
6	Instrumentalness	0.052132
12	Self-Reflection and Personal Struggles	0.040318
2	Key	0.037704
14	Celebration and Fun	0.018412
20	Fun and Humor	0.017693
11	Love and Relationships	0.011462
16	Storytelling and Narrative	0.007339
17	Escape and Fantasy	0.004930
18	Spiritual and Religious	0.004881
13	Social and Political Themes	0.004166
19	Cultural and Lifestyle	0.004068
15	Philosophical and Existential	0.002923

Metrics and insights: XGBoost

Model of Choice: XGBosst with MultiOutputClassifier

- **Boosting Technique:** Uses gradient boosting to improve performance.
- **Feature Importance:** Identifies key contributing factors in classification.
- **Efficiency:** Optimized for speed and accuracy, handles large datasets well.

Interpretation of Results:

- **Test Accuracy (0.2012):** Slightly higher than Random Forest, but still reflects the challenge of exact label matching.

- **Micro F1-Score (0.54):** Shows a small improvement over Random Forest (0.5294), indicating better balance between precision and recall across all labels.

Comparison to Random Forest:

- **Slightly better Micro F1-Score**, suggesting improved classification consistency.
- **Higher accuracy**, but the overall challenge of multi-label classification remains.
- **Potential Trade-offs:** XGBoost may be more computationally intensive compared to Random Forest

Summary of findings:

1. We were able to find
   a. meaningful clusters in the audience-response data to the songs we scraped for.
   b. meaningful clusters for the songs based on spotify meta-data.

2. The clusters can be relied upon to retrieve quality recommendations using distance metrics to evaluate similarity.

3. We also tried to find a predictive relationship between spotify metadata (in conjunction with song lyric topics) and types of audience response. While our results weren’t perfect, given the complexity of the dataset, the results can be considered reasonable.

Conclusion

Project Summary:

- Developed a recommendation system using **audio features** (e.g., danceability, tempo, energy), **user-generated comments**, and **lyric data**.
- Applied **multi-label** and **multi-class classification** techniques to assign multiple emotions (multi-label) and primary emotional tones (multi-class) to each song.

Key Takeaways:

- Combining **audio features**, **user-generated comments**, and **lyric data** provides a powerful framework for music recommendation.
- Using both **multi-label** and **multi-class classification** enables the system to offer diverse and personalized suggestions.

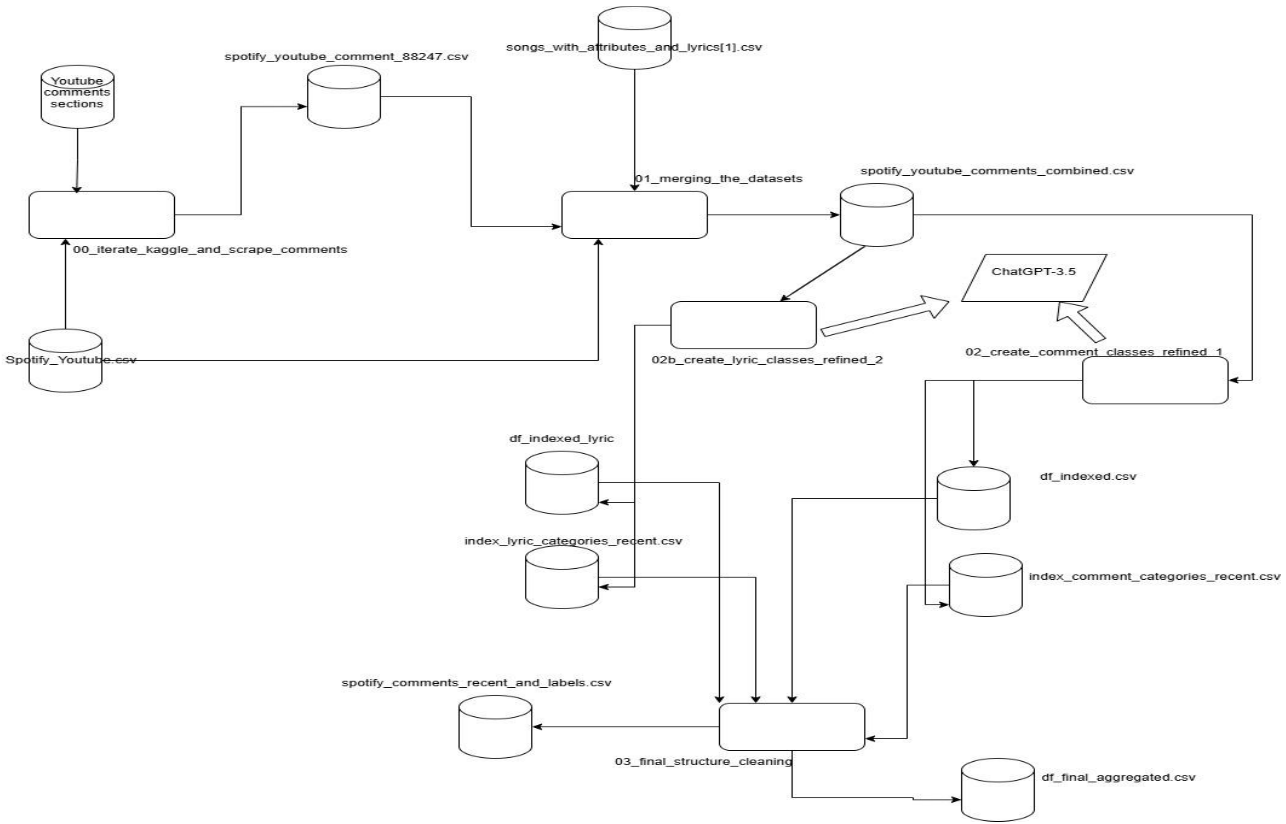
Some Challenges:

1. Dataset preparation involved web scraping and LLM prompting.
2. Finding meaningful clusters.
3. LLM prompting to generate meaningful labels.
    a. Manual inspection to assess quality
4. Multilabel multiclass classification model improvements.

Appendix A: References

- Spotify URL & YouTube URL: https://www.kaggle.com/datasets/salvatorerastelli/spotify-and-youtube/data
- Spotify song attributes & lyrics: https://www.kaggle.com/datasets/bwandowando/spotify-songs-with-attributes-and-lyrics

Appendix B: ETL diagram for dataset preparation

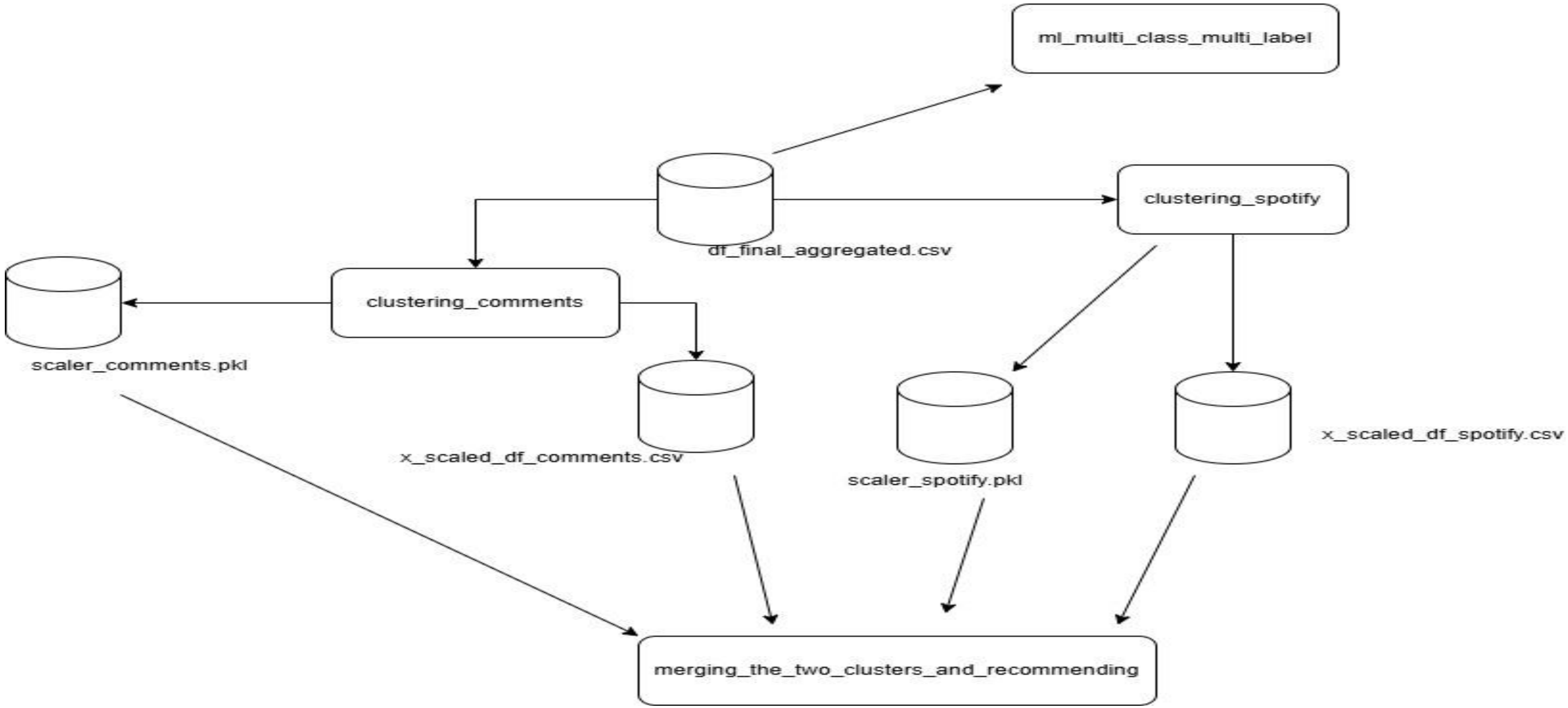


Appendix C : Dataset and process descriptions:

- **spotify_Youtube.csv:** youtube URL, corresponding Spotify-ID, spotify meta-data.
- **spotify_youtube_comment_88247.csv:** SpotifyID, youtube video ID, comments and likes.
- **songs_with_attributes_and_lyrics[1].csv:** spotify id**,** lyrics.
- **spotify_youtube_comments_combined.csv:** spotify_id, video_id, comment, likes, spotify meta-data, lyrics
- **df_indexed.csv:** indexed records for which comment labels were successfully generated.
- **df_indexed_lyric.csv:** indexed records for which lyric labels were successfully generated.
- **index_lyric_categories_recent.csv:** indexed records, spotify id, lyric labels
- **index_comment_categories_recent.csv:** indexed records, spotify id, comment labels
- **spotify_comments_recent_and_labels.csv:** spotify id, meta-data, comment-labels
- **df_final_aggregated.csv:** spotify id, meta-data, comment-category counts, lyric labels

- **00_iterate_kaggle_and_scrape_comments.ipynb:** iterate over records in **spotify_youtube.csv** and generate **spotify_youtube_comment_88247.csv** by leveraging youtube API.
- **01_merging_the_datasets.ipynb:** Generating **spotify_youtube_comments_combined.csv** by merging with **songs_with_attributes_and_lyrics[1].csv**
- **02_create_comment_classes_refined_1.ipynb:** iterate over **spotify_youtube_comments_combined.csv** and generate **df_indexed.csv** and **index_comment_categories_recent.csv** by prompting chatGPT.
- **02b_create_lyric_classes_refined_2.ipynb:** iterate through **spotify_youtube_comments_combined.csv** and generate **df_indexed_lyric.csv** and **index_lyric_categories_recent.csv** by prompting chatGPT.
- **03_final_structure_cleaning.ipynb:** combine and clean all the datasets to generate **df_final_aggregated.csv.**

Appendix D: Workflow for ML and Data Mining



Appendix E:Dataset and process descriptions

- **x_scaled_df_comments.csv**: Spotify Id, comment quantities scaled, cluster labels.
- **x_scaled_df_spotify.csv**: Spotify Id, spotify metadata quantities scaled, cluster labels.

- **clustering_comments.ipynb**: applying clustering algorithms on top of comments data
- **clustering_spotify.ipynb**: applying clustering algorithms on top of spotify data
- **ml_multi_class_multi_label.ipynb**: trying to discover predictive relationships between spotify metadata, lyric labels and comment-labels.

Appendix F: YouTube Comments Category

YouTube Comment Code Snippet:

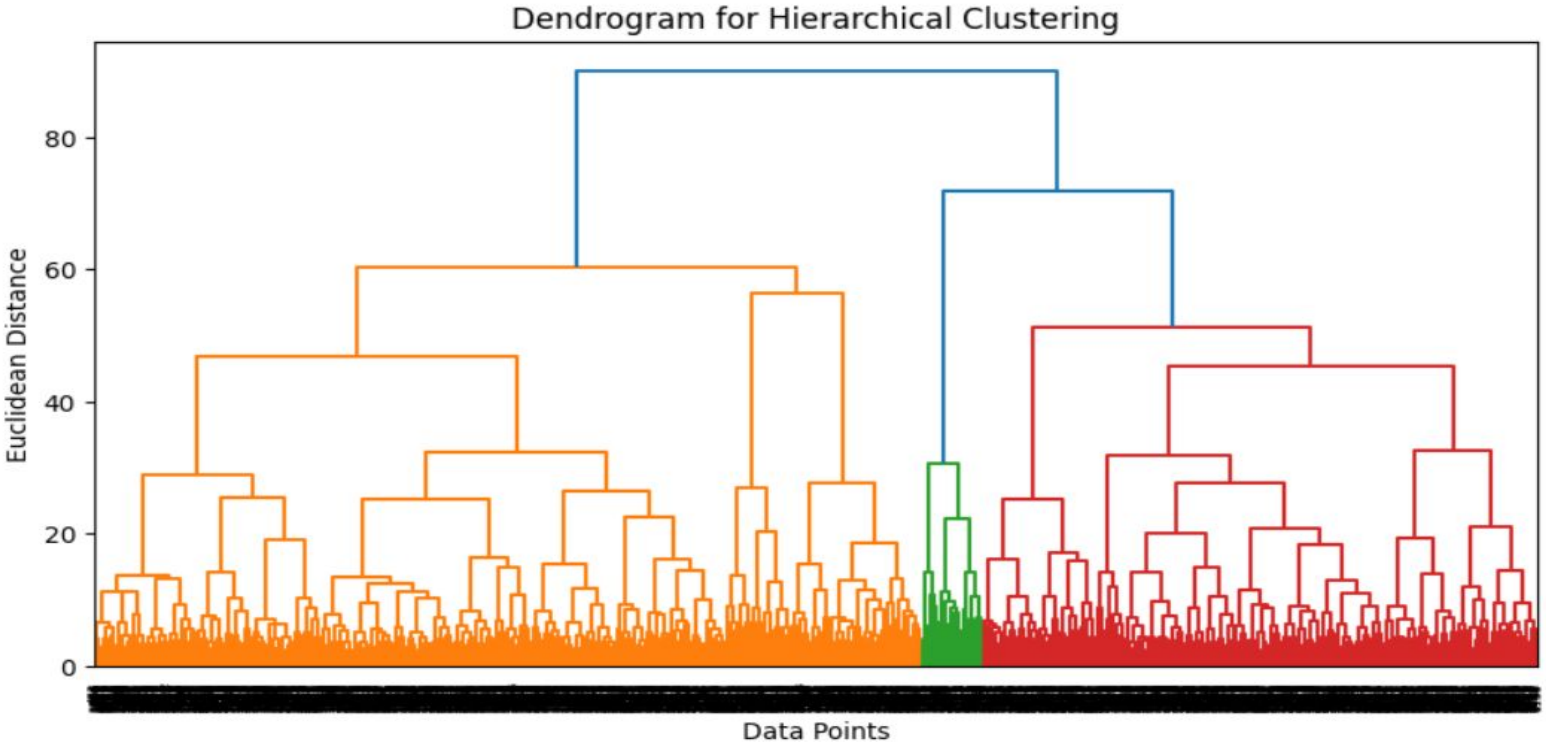
```
youtube_comment_categories = [
    "Humor and Memes",
    "Appreciation and Praise",
    "Words of encouragement",
    "Words of empathy",
    "Personal Stories and Experiences",
    "Nostalgia and Memories"
]
```

Appendix G: Lyrics Topic Modeling

Lyrics Topic Modeling Code Snippet

```
youtube_lyric_categories = [
    "Love and Relationships", "Self-Reflection and Personal Struggles", "Social and Political Themes",  "Celebration and Fun",
    "Philosophical and Existential", "Storytelling and Narrative", "Escape and Fantasy", "Spiritual and Religious",
    "Cultural and Lifestyle", "Fun and Humor"
]
```

Appendix H: Spotify Data Clustering



Appendix J: Library/ Frameworks used

- openai
- pandas
- sklearn
- matplotlib